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Introduction

For almost twenty years, I had been holding on to an old ham radio magazine that contains an article describing the construction of a Lyonodyne style crystal radio. Finally, in 2004 I got around to building it.

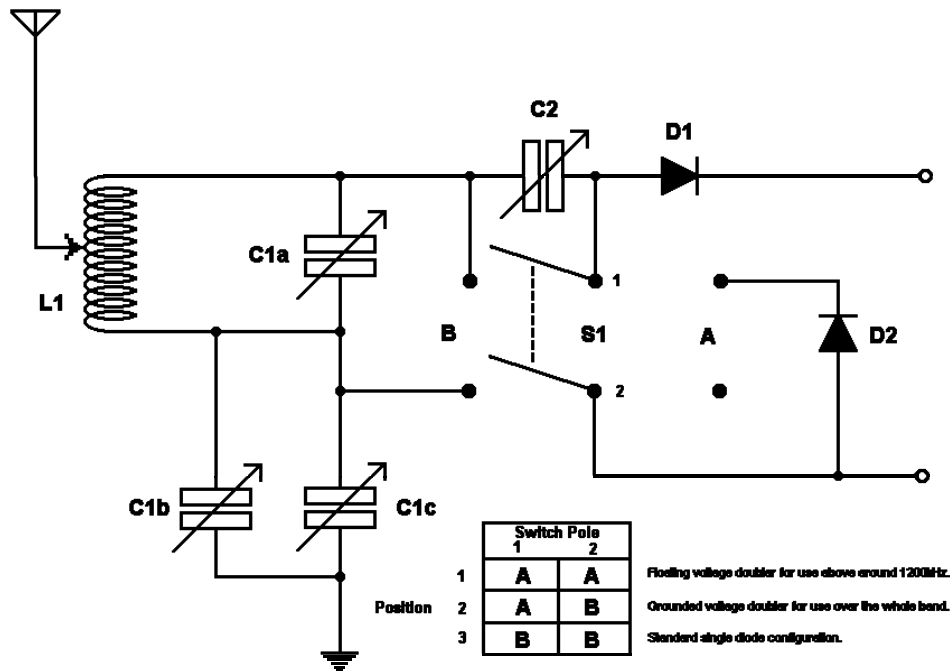
Further reading

Based on an [article](#) published in "Amateur Radio Action", Vol. 4 No. 13 - p29-33, - 27th April 1981.

From an original article by R.M. Tuggle published in "DX News", bulletin of the National Radio Club Vol. 44, No. 25, p 5-8, 23rd May 1977.

Construction

A 3 gang capacitor is called for in this circuit. One gang is used to resonate L1, while the remaining two gangs provide capacitive coupling to ground. All three gangs are around 365pf. You could try and make do with a dual gang unit, (i.e. only 365pf max to ground from the tank coil.) but I found that there was a loss in sensitivity at the low frequency end of the band with that arrangement. In a pinch, you could probably use one dual gang from tank coil to ground and one half of another dual gang for the tank itself. Tuning will be a little difficult but not impossible. The gang I used came from an old MW/SW radio.



The coil is 30 turns of 1mm enameled copper wire wound on an 11 slot rook former (one over, two under) about 120mm in diameter. It measures 200uH. It's tapped every sixth turn. The coil is held together using wax lacing string and is held in place with glue from a hot glue gun.

I used a pair of OA90 diodes in a special switching arrangement that allows for three detector configurations.

The first setting is a floating voltage doubler with high selectivity for use above around 1200 kHz.

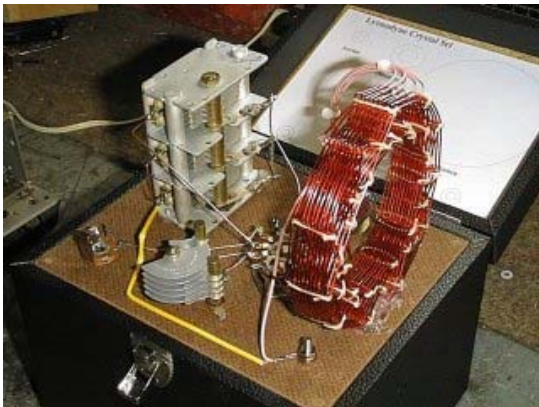
Position two is a grounded voltage doubler for use over the whole band.

Position three is a standard single diode configuration which is less selective but has more volume than the other two positions.

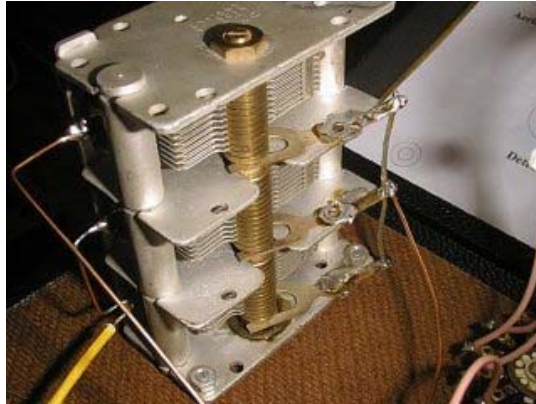
The trimmer cap (C2) that feeds the detectors is used as a selectivity control when switch S1 is in positions one or two. A small value variable unit (30pf max) is used here.

I built the set into a nice vinyl covered box that I found in an Army Disposal store.

The complete set fits into a (170mm x 215mm x 175mm HxWxD) box with a hinged lid.



As always, good mechanical/electrical connections are a must with these sets. The gang I used does not rely on the connection through the bearings for the rotor. Rather it uses silver fingers to connect to the rotor shaft. Note that each finger is soldered to the frame of the capacitor making a good connection.



Headphones.

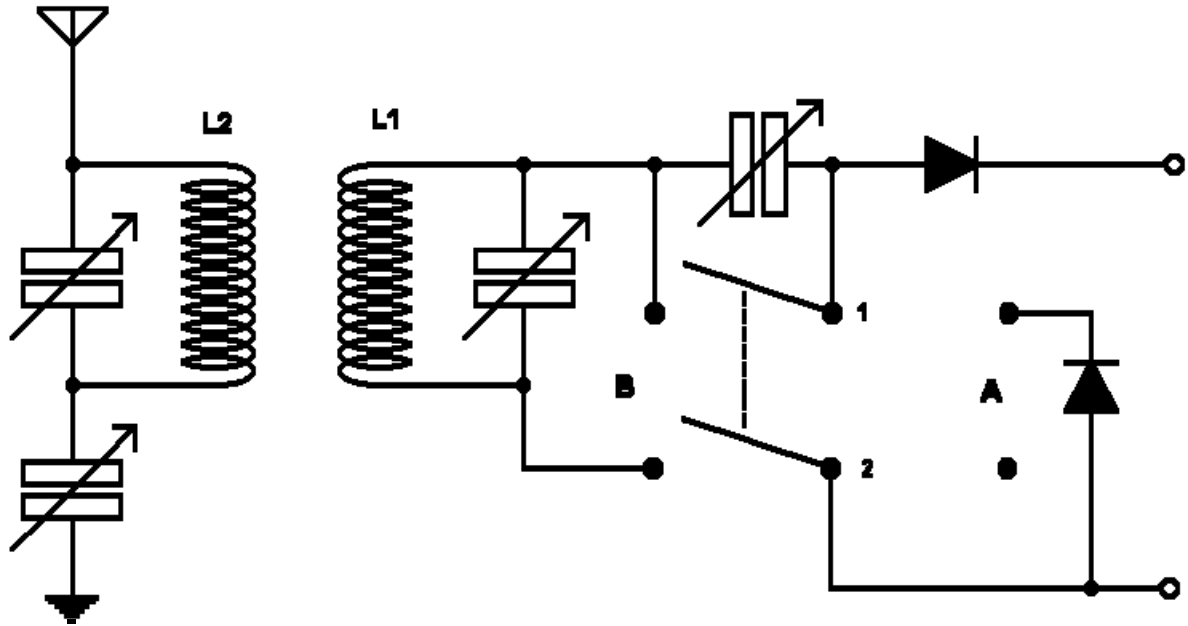
The original 1940's style bakelite headphones I used measured around 4000 ohms DC. These sounded quite good and the set performed well but I felt it could be better. I ended up using a set of very sensitive military 'phones coupled to the set via a high quality audio transformer with a multi-tapped primary from 30 ohms to 1000 ohms and a 4800 ohm (160 ohm DC) secondary. I mounted the transformer in a utility box and drove the crystal set into the secondary and selected the appropriate tap from the primary to drive the headphones.

It really is worth the effort to try and match the output of your crystal set to your headphones.



Loose Coupling.

I followed up with a second identical Rook coil connected in Lyonodyne configuration loose coupled to the original set. The antenna and ground now connect to this antenna coil and the set now acts as a tuned detector coil feeding the headphones. The best detector configuration is in switch position 3.



L2 is a copy of L1 but is tuned by a dual gang 365pf capacitor. The two sections of the capacitor that were connected to ground from L1 have been removed from the diagram for clarity but still exist. Because they are no longer connected to ground they play very little if any part in the operation of the set. Coupling between the two coils is adjusted by moving the two coils closer or further apart. The closer they are, the louder the signals in the headphones. There comes a point however, where moving the coils closer together results in a decrease in sensitivity and selectivity. The trick is to find this point where the coupling between the two coils is optimum. For my set, this distance is one coil diameter (200mm). I mounted L2 in a matching vinyl box.





The change in performance brought about by going to loose coupling with a tuned detector coil AND the addition of a quality audio transformer and sensitive headphones have been impressive. This level of improvement makes it easier to use a crystal radio in the city even when you're surrounded by high power stations.